

UUIC Technical Report: Multi-Organ Classification and Segmentation via Prompt-Guided Hyper-Perception with UPerNet Refinement and Uncertainty-Aware Loss

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Abstract. We present our solution for the Universal Ultrasound Image Challenge (UUIC), targeting multi-organ classification and segmentation under heterogeneous acquisition conditions. Our method builds upon the Prompt-Guided Hyper-Perception and Attention-Matching Downstream Synchronization framework, with three key modifications: *(i)* a unified 4-class classification head with task-adaptive channel selection; *(ii)* integration of UPerNet after the Swin Transformer encoder for fine-grained segmentation refinement; and *(iii)* uncertainty-weighted loss combining aleatoric and epistemic components, together with a KL divergence term to regularize the predicted distribution. Experimental results on the official validation set demonstrate competitive segmentation accuracy, robust classification performance, and efficient memory usage, highlighting the method’s applicability to real-world ultrasound workflows.

Keywords: Ultrasound imaging · Multi-task learning · Prompt-guided learning · UPerNet · Uncertainty estimation

1 Introduction

The Universal Ultrasound Image Challenge (UUIC) [1] evaluates algorithms capable of performing **multi-organ, multi-disease classification and segmentation** using diverse, real-world ultrasound data collected from multiple centers and devices [1]. Each target organ is associated with a distinct clinical task, including nodule malignancy classification (breast), organ contour delineation (kidney, cardiac), and disease diagnosis (appendix, fatty liver).

The challenge assesses segmentation via Dice Similarity Coefficient (DSC) and Normalized Surface Distance (NSD), classification via Area Under the ROC Curve (AUC) and accuracy, and efficiency via inference time and maximum GPU memory usage.

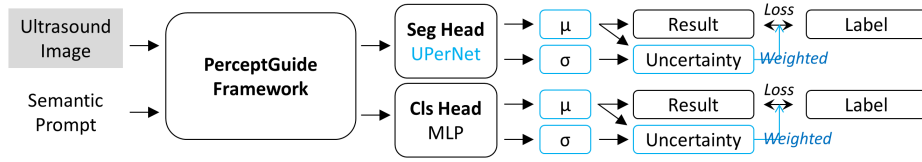


Fig. 1. Overview of our framework.

We base our approach on the PerceptGuide orchestration framework [2], which unifies classification and segmentation through semantic prompts. However, we observed three limitations in the original framework:

- The classification head only supports binary classification, limiting flexibility for tasks with more than two classes.
- The segmentation head may lack sufficient granularity for boundary refinement.
- The training loss equally treats all samples, potentially underweighting hard cases with high uncertainty.

We address these by introducing a unified 4-class classification head, UPerNet refinement for segmentation, and uncertainty-weighted loss with KL divergence regularization.

2 Method

2.1 Multi-task Paradigm

The baseline orchestration framework [2] jointly addresses classification and segmentation through a shared Swin Transformer encoder guided by semantic prompts (*Object*, *Task*, *Input*, and *Position*). While this design ensures feature sharing, its original binary classification head and lightweight segmentation branch limit flexibility for multi-class tasks and fine-grained boundary delineation.

To address these issues, we adopt a unified **multi-task paradigm** (Fig. 1) with the following changes:

- **Unified 4-class classification head:** Replaces the original binary head with a 4-class linear classifier, trained with multi-class cross-entropy. This allows a single model to support both binary and multi-class classification without re-training. At inference, we select the relevant output channels according to the task definition (first two for binary, all for four-class tasks).
- **Task-adaptive output selection:** A lightweight routing mechanism ensures that only the necessary classification channels are activated per task, avoiding interference from irrelevant outputs.
- **Prompt-aware feature sharing:** Classification and segmentation tasks share the same encoder features, ensuring that prompt conditioning benefits both tasks.

This paradigm retains the efficiency of the baseline orchestration design while extending its flexibility to heterogeneous task definitions in UUIC.

2.2 Model Architecture

The shared Swin encoder outputs $B \times 96 \times 224 \times 224$ features. Our two heads run in parallel:

UPerNet-based segmentation head. We concatenate the encoder feature with the input to obtain a $B \times 99 \times 224 \times 224$ tensor and feed it to UPerNet for refined masks. *Crucially*, the segmentation head **co-predicts** both the mask logits and $(\mu_{\text{seg}}, \log \sigma_{\text{seg}}^2)$ (per-pixel), enabling aleatoric estimation within the same forward pass.

Lightweight 4-class classification head. Global average pooling followed by a fully-connected layer produces the logits $\hat{y}_{\text{cls}} \in \mathbb{R}^{B \times 4}$; at inference we select channels according to the task (first two for binary; all four for 4-class). The classification head also **co-predicts** $(\mu_{\text{cls}}, \log \sigma_{\text{cls}}^2)$ (per-class).

Compared to the baseline orchestration framework, our design integrates a stronger decoder (UPerNet), a flexible classifier (4-class with task-adaptive routing), and *co-head uncertainty prediction* without introducing an extra branch.

2.3 Loss Function

We jointly model *aleatoric* uncertainty u_{alea} (network-predicted σ from $(\mu, \log \sigma^2)$) and *epistemic* uncertainty u_{epi} (variance across S MC-dropout passes). A KL regularization aligns (μ, σ) with batch-level (μ_B, σ_B) . Task losses are CE for classification and a CE–Dice combination for segmentation ($\lambda_1=0.4$, $\lambda_2=0.6$). Uncertainty weights are

$$w_{\text{alea}} = \frac{1}{1 + 0.5 u_{\text{alea}}}, \quad w_{\text{epi}} = 1 + 0.5 u_{\text{epi}}.$$

The final loss is

$$L_{\text{final}} = (w_{\text{alea}} w_{\text{epi}}) L_{\text{task}} + \lambda_{\text{KL}} \overline{D}_{\text{KL}}, \quad \lambda_{\text{KL}} = 0.001.$$

3 Experimental Results

3.1 Implementation Details

- **Framework:** PyTorch 1.12
- **Batch size:** 32
- **Learning rate:** 1×10^{-4} , Adam optimizer
- **Hardware:** NVIDIA RTX 3070 GPU
- **Epochs:** 200
- **Data policy:** only UUIC-provided datasets; public pre-trained models allowed

Table 1. Performance on UUIC validation set. Segmentation: DSC / NSD; Classification: Accuracy / AUC. GPU memory measured during inference.

Task	Accuracy	AUC	DSC	NSD
Appendix (cls)	0.653	0.691	–	–
Breast (cls)	0.653	0.720	–	–
Breast Luminal (cls)	0.496	0.570	–	–
Liver (cls)	0.802	0.843	–	–
Breast (seg)	–	–	0.845	0.205
Breast Luminal (seg)	–	–	0.829	0.119
Cardiac (seg)	–	–	0.840	0.060
Fetal Head (seg)	–	–	0.928	0.102
Kidney (seg)	–	–	0.887	0.111
Thyroid (seg)	–	–	0.617	0.072

3.2 Validation Results

Maximum GPU memory usage is approximately 2.2 GB. Official running time evaluation is pending.

4 Discussion and Conclusion

The unified 4-class classification head enables task-adaptive inference without retraining, UPerNet refines boundaries by leveraging low-level image cues, and uncertainty-weighted loss with KL regularization emphasizes difficult samples. This yields robust segmentation on challenging organs and strong classification for liver and appendix tasks, while maintaining low inference memory usage. Performance on fine-grained breast subtype classification remains an area for improvement.

Future work will focus on multi-scale feature fusion for classification and dynamic prompt selection for further generalization.

References

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